UK e-Commerce
Customer Segmentation

Daud Rusyad Nurdin

Introduction

My name is Daud Rusyad Nurdin (Daud). Coming from Bandung, I’m an IT practitioner in Jakarta and live in Tangerang Selatan. I used to play futsal and badminton, but retired. Reading any kind of topics book and learning something new is my hobby. Maybe because I think I’m a curious man.

Graduated my bachelor in Informatics (Information Technology), then while I was working I continued my magister program in Management, with a specialization in International Business.

Starting out as a programmer, then I've worked extensively in data processing and have been involved in several data warehouse and business intelligence projects. I'm currently interested in data science, and I plan to delve deeper into AI programming and Generative AI. My colleagues say I'm a very detailed and meticulous person. My work principle: correct, fast, and neat.

Project:
UK e-commerce
Customer Segmentation

- Project Background
- Business Problem
- Data Understanding
- Data Analysis and Insights
- Dashboards and Visualizations
- Recommendations and Actionable Insights

Project Background

- From the past to the present, customers remain a crucial factor in the success of any business. And the motto "The customer is king" remains relevant today.
- This project will analyze UK e-commerce transaction data to identify consumer segmentation based on several categories within the data.
- RFM analysis will be performed on this data to identify customer segments using this popular and widely used method. This allows for accurate customer segmentation and provides recommendations & actionable insights for each segment.
- With the recommendations provided, it is hoped that the level of customer satisfaction and customer retention can be increased and "lost" customers can be brought back.

Business Problem

- In today's competitive business environment, customers are a company's most valuable asset. The principle "The customer is king" emphasizes that customers play a central role in determining a company's success. By understanding and meeting customer needs, companies can increase customer satisfaction and build strong loyalty.
- From Balanced Scorecard perspective, customer satisfaction is a crucial aspect in measuring company performance. By prioritizing customer satisfaction, companies can increase revenue, retain customers, and enhance their reputation.


The Customer is King

Data Understanding

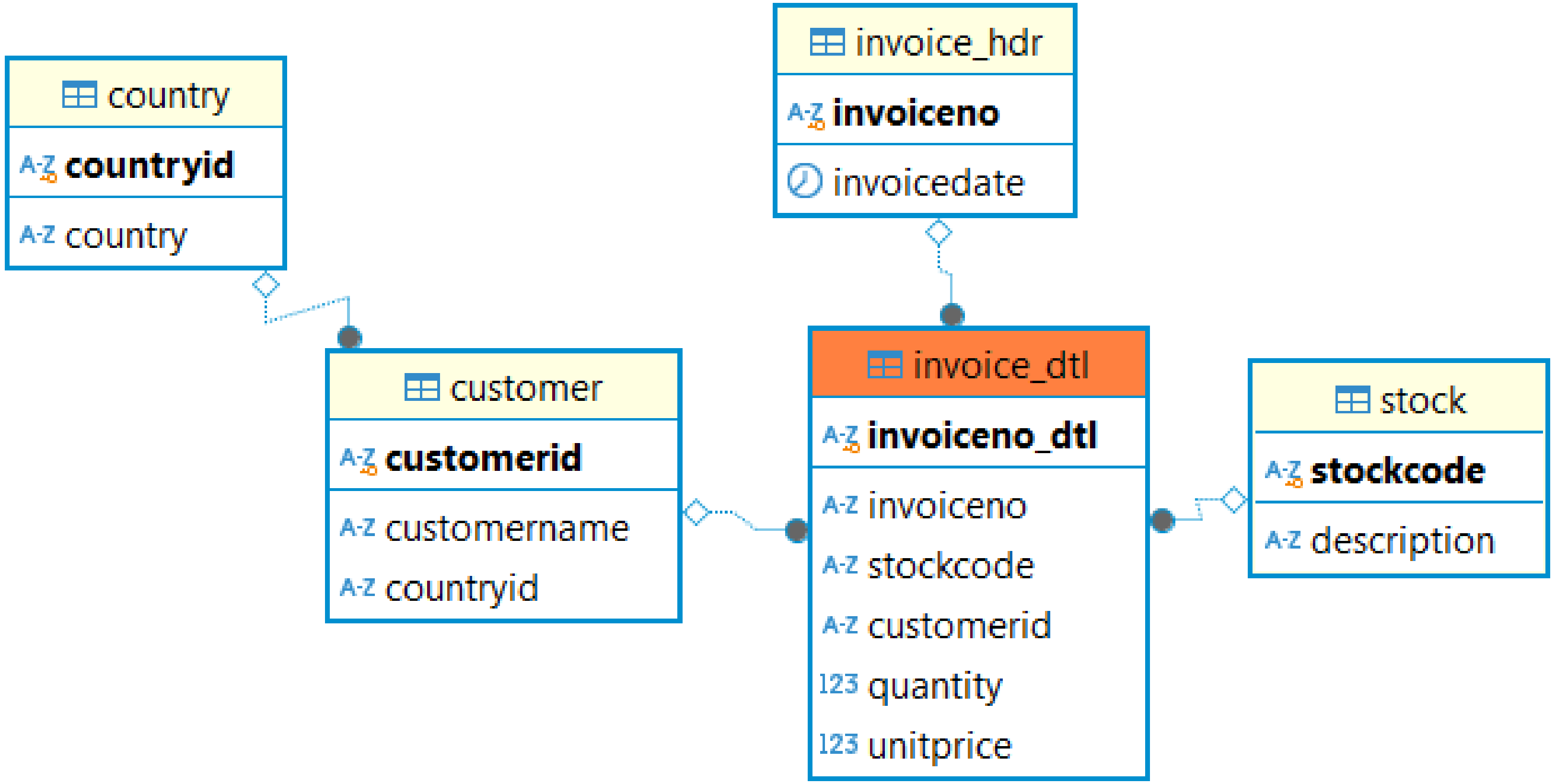
Data Source and its Structure

- Data Source: uk_ecommerce.csv*)
- Data structure:

No	Column Name	Data Type	Description
1	InvoiceNo	String	Every bill generated of various products has a unique Invoice No. --> Primary Key of Invoice
2	StockCode	String	Every product has a unique stock code --> Primary Key
3	Description	String	Description of product
4	Quantity	Integer	Amount of product bought
5	InvoiceDate	Date	The datetime when bill was generated
6	UnitPrice	Float	Price of product (Unit price)
7	CustomerID	String	Every account has a unique id --> Primary key of
8	Country	String	Country of customer

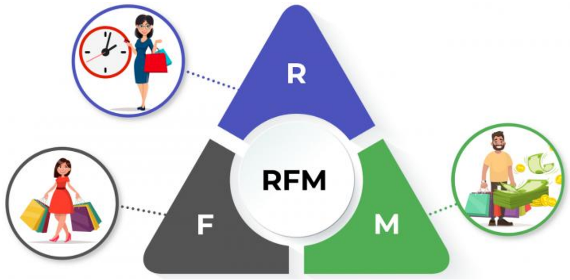
*) https://www.kaggle.com/code/calvinsinaalkhazhaj/eda-uk-ecommerce-analysis

Entity Relationship Diagram



Data Sample

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	17850.00	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	17850.00	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	17850.00	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/1/2010 8:26	3.39	17850.00	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/1/2010 8:26	3.39	17850.00	United Kingdom



Data Cleansing

Global information of data

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
 #   Column       Non-Null Count   Dtype  
---  ------       --------------   -----  
 0   InvoiceNo    541909 non-null  object 
 1   StockCode    541909 non-null  object 
 2   Description  540455 non-null  object 
 3   Quantity     541909 non-null  int64  
 4   InvoiceDate  541909 non-null  object 
 5   UnitPrice    541909 non-null  float64
 6   CustomerID   406829 non-null  float64
 7   Country      541909 non-null  object 
dtypes: float64(2), int64(1), object(5)
memory usage: 33.1+ MB
```

<u>Information obtained</u>:

1. Total record: 541,909 rows; total column: 8 columns.
2. There are two columns that contain Null values, namely: (1) Description; and (2) CustomerID → Data cleaning is required.
3. It would be better to transform data type of CustomerID *from float to object*.

Change data type of CustomerID

Before

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	17850.00	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	17850.00	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	17850.00	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/1/2010 8:26	3.39	17850.00	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/1/2010 8:26	3.39	17850.00	United Kingdom

After

```
'''
  Convert data from float to string:
  (1) change data type from string to integer to eliminate decimal point
  (2) change data type from integer to string
'''
df['CustomerID'] = pd.to_numeric(df['CustomerID'], errors='coerce').astype('Int64') # change null to nan
df['CustomerID'] = df['CustomerID'].apply(lambda x: 'null' if pd.isnull(x) else str(int(x)))
df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	17850	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	17850	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	17850	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/1/2010 8:26	3.39	17850	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/1/2010 8:26	3.39	17850	United Kingdom

```
df.dtypes
```

	0
InvoiceNo	object
StockCode	object
Description	object
Quantity	int64
InvoiceDate	object
UnitPrice	float64
CustomerID	object
Country	object

dtype: object

Data Duplicate

```
# Check data duplicate
dupl = df.duplicated().sum()
print(f"Total duplicate: {dupl}")
```

```
Total duplicate: 5268
```

```
# Find sample of duplicate
df_dup = df[df.duplicated(keep=False)]
df_dup = df_dup.sort_values(by=['StockCode', 'CustomerID'])
df_dup.head(10)
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
139206	548309	10120	DOGGY RUBBER	1	3/30/2011 12:02	0.21	16657	United Kingdom
139262	548309	10120	DOGGY RUBBER	1	3/30/2011 12:02	0.21	16657	United Kingdom
248501	558860	10125	MINI FUNKY DESIGN TAPES	1	7/4/2011 12:18	0.85	17975	United Kingdom
248503	558860	10125	MINI FUNKY DESIGN TAPES	1	7/4/2011 12:18	0.85	17975	United Kingdom
239410	558049	10133	COLOURING PENCILS BROWN TUBE	1	6/24/2011 14:16	0.42	15850	United Kingdom
239419	558049	10133	COLOURING PENCILS BROWN TUBE	1	6/24/2011 14:16	0.42	15850	United Kingdom
308696	564049	10133	COLOURING PENCILS BROWN TUBE	10	8/22/2011 13:30	0.42	17585	United Kingdom
308705	564049	10133	COLOURING PENCILS BROWN TUBE	10	8/22/2011 13:30	0.42	17585	United Kingdom
396050	571054	10135	COLOURING PENCILS BROWN TUBE	1	10/13/2011 13:58	1.25	14234	United Kingdom
396055	571054	10135	COLOURING PENCILS BROWN TUBE	1	10/13/2011 13:58	1.25	14234	United Kingdom

Total duplicate: 5268 rows
→ Need to deduplicate

Remove Duplicate

Before

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
139206	548309	10120	DOGGY RUBBER	1	3/30/2011 12:02	0.21	16657.00	United Kingdom
139262	548309	10120	DOGGY RUBBER	1	3/30/2011 12:02	0.21	16657.00	United Kingdom

```
'''
  Lakukan proses deduplikasi disini...
'''
totb4dedup = df.shape[0] # catat tota record sebelum deduplikasi

df.drop_duplicates(keep='first', inplace=True) # remove duplicate

print(f"Total data before deduplicate: {totb4dedup}")
print(f"Total duplicate data          : {dupl}") # telah dihitung sebelumnya
print(f"Total data after deduplicate : {df.shape[0]}")
```

```
Total data before deduplicate: 541909
Total duplicate data         : 5268
Total data after deduplicate : 536641
```

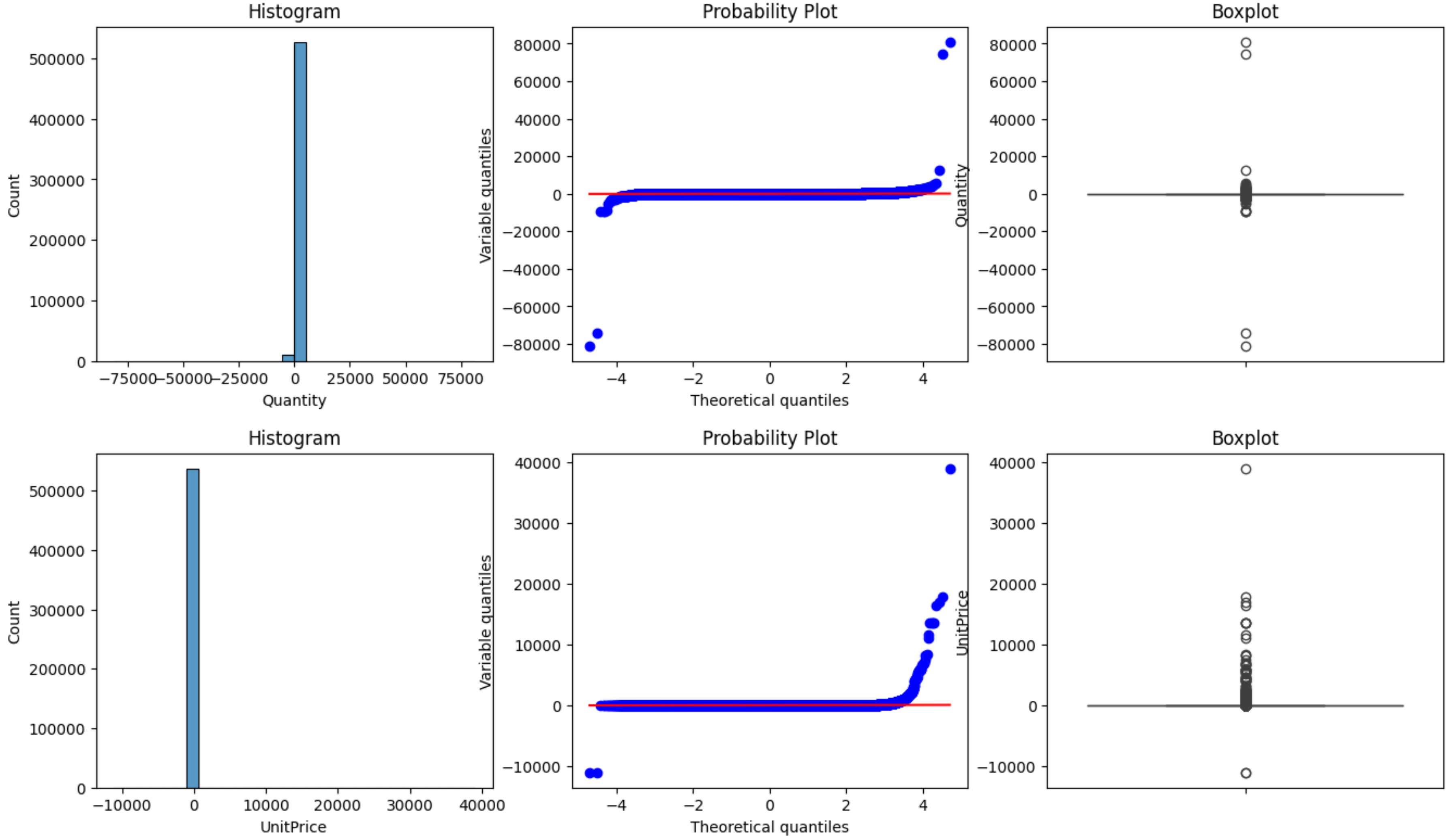
After

```
df.query('InvoiceNo == "548309" and StockCode == "10120"')
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
139206	548309	10120	DOGGY RUBBER	1	3/30/2011 12:02	0.21	16657	United Kingdom

After removing duplicate data, there are 536,641 rows of data left.

Outliers



Before

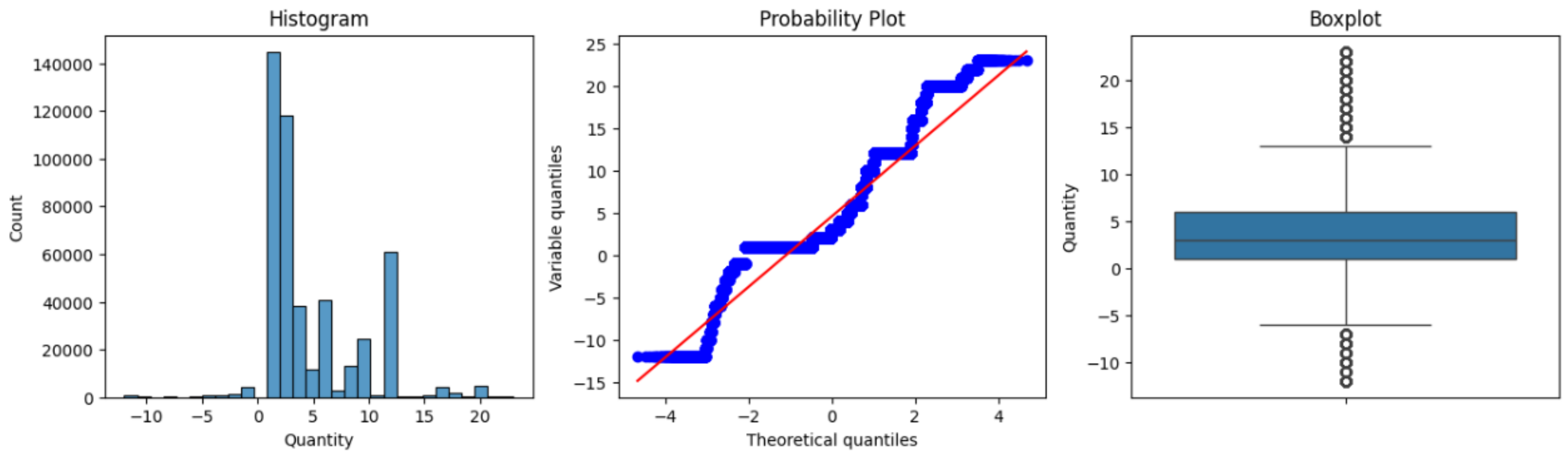
```
[143] # Descriftive statistics
      df.describe().T
```

	count	mean	std	min	25%	50%	75%	max
Quantity	536641.00	9.62	219.13	-80995.00	1.00	3.00	10.00	80995.00
UnitPrice	536641.00	4.63	97.23	-11062.06	1.25	2.08	4.13	38970.00

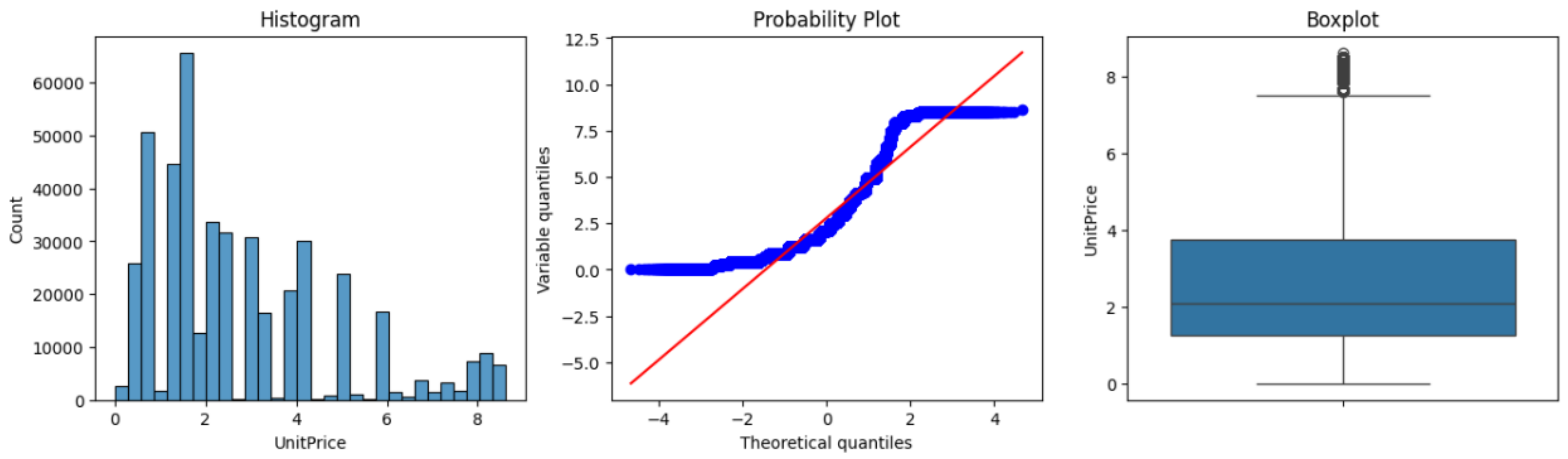
Remove outliers

Remove Outliers

```
Kolom: Quantity, boundary (lower - upper): -12.5000 to  23.5000
```




```
Kolom: UnitPrice, boundary (lower - upper): -3.1900 to  8.6500
```



After

```
# Descriftive statistics
df.describe().T
```

	count	mean	std	min	25%	50%	75%	max
Quantity	445792.00	4.80	4.61	-12.00	1.00	3.00	8.00	23.00
UnitPrice	445792.00	2.79	2.04	0.00	1.25	2.10	3.75	8.62

After removing outliers, there are 445,792 rows of data left.

Negative value of Quantity and UnitPrice

```
def_neg = df.query('Quantity < 0 or UnitPrice < 0')
def_neg.shape
```

(10626, 8)

```
[29] df.query('Quantity < 0')
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
141	C536379	D	Discount	-1	12/1/2010 9:41	27.50	14527.00	United Kingdom
154	C536383	35004C	SET OF 3 COLOURED FLYING DUCKS	-1	12/1/2010 9:49	4.65	15311.00	United Kingdom
235	C536391	22556	PLASTERS IN TIN CIRCUS PARADE	-12	12/1/2010 10:24	1.65	17548.00	United Kingdom
236	C536391	21984	PACK OF 12 PINK PAISLEY TISSUES	-24	12/1/2010 10:24	0.29	17548.00	United Kingdom
237	C536391	21983	PACK OF 12 BLUE PAISLEY TISSUES	-24	12/1/2010 10:24	0.29	17548.00	United Kingdom
...	...	...	...	...	...	...	...	...
540449	C581490	23144	ZINC T-LIGHT HOLDER STARS SMALL	-11	12/9/2011 9:57	0.83	14397.00	United Kingdom
541541	C581499	M	Manual	-1	12/9/2011 10:28	224.69	15498.00	United Kingdom
541715	C581568	21258	VICTORIAN SEWING BOX LARGE	-5	12/9/2011 11:57	10.95	15311.00	United Kingdom
541716	C581569	84978	HANGING HEART JAR T-LIGHT HOLDER	-1	12/9/2011 11:58	1.25	17315.00	United Kingdom
541717	C581569	20979	36 PENCILS TUBE RED RETROSPOT	-5	12/9/2011 11:58	1.25	17315.00	United Kingdom

10624 rows × 8 columns

```
df.query('UnitPrice < 0')
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
299983	A563186	B	Adjust bad debt	1	8/12/2011 14:51	-11062.06	NaN	United Kingdom
299984	A563187	B	Adjust bad debt	1	8/12/2011 14:52	-11062.06	NaN	United Kingdom

Negative values in Quantity and UnitPrice are present, likely due to cancelled transactions. *)

Remove the data
→ not relevant

*) https://www.kaggle.com/code/duygujones/customer-behaviour-analysis-eda-rfm-pca

Remove cancelled transaction

```
'''
  Buang data dengan Quantity negatif & UnitPrice negatif
'''
totrecb4 = df.shape[0]
df = df[~((df['Quantity'] < 0) | (df['UnitPrice'] < 0))]

print(f'Total record before cleansing: {totrecb4}')
print(f'Total record negative removed: {df_neg.shape[0]}')
print(f'Total record after cleansing : {df.shape[0]}')
```

```
Total record before cleansing: 445792
Total record negative removed: 7222
Total record after cleansing : 438570
```

Anomaly Data

*) Data Sample

StockCode	Description
10002	INFLATABLE POLITICAL GLOBE
10002	
10080	GROOVY CACTUS INFLATABLE
10080	check
10080	
10120	DOGGY RUBBER
10123C	HEARTS WRAPPING TAPE
10123C	
10123G	
10124A	SPOTS ON RED BOOKCOVER TAPE
10124G	ARMY CAMO BOOKCOVER TAPE
10125	MINI FUNKY DESIGN TAPES
10133	damaged
10133	COLOURING PENCILS BROWN TUBE

StockCode	Description
84795b	SUNSET CHECK HAMMOCK
84796A	PINK HAWAIIAN PICNIC HAMPER FOR 2
84796B	
84796B	BLUE SAVANNAH PICNIC HAMPER FOR 2
84796a	PINK HAWAIIAN PICNIC HAMPER FOR 2
84797B	
84798A	PINK FOXGLOVE ARTIIFCIAL FLOWER

StockCode	Description
Every product has a unique stock code	Info. of the product
4070 unique values	**4224** unique values

There is no constraint validation in the source data database table.

```
[281] df.query('Description == "damaged"')
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
154799	549948	37488A	damaged	-1	4/13/2011 12:55	0.00	NaN	United Kingdom
166836	550954	47013A	damaged	-2	4/21/2011 16:18	0.00	NaN	United Kingdom
200352	554128	84563A	damaged	-4	5/23/2011 9:26	0.00	NaN	United Kingdom
212679	555503	84313C	damaged	-1	6/3/2011 15:39	0.00	NaN	United Kingdom

Transform StockCode → Use function Upper()

Anomaly Data

Description	Description
20713	stock check
?	stock creditted wrongly
? sold as sets?	taig adjust
??	taig adjust no stock
?? missing	temp adjustment
???	test
????damages????	throw away
????missing	thrown away
???lost	Thrown away.
???missing	thrown away-can't sell
?display?	thrown away-can't sell.
?lost	Thrown away-rusty
?missing	to push order througha s stock was
?sold as sets?	Unsaleable, destroyed.
add stock to allocate online orders	water damage
adjust	Water damaged
Adjust bad debt	website fixed
Adjustment	wet
alan hodge cant mamage this section	wet boxes
allocate stock for dotcom orders ta	wet damaged

From the description information it seems that this data is not real sales transaction data.

This type of data will be excluded from the dataset.

Remove this data.

Remove anomaly data

Methodology:

- Identify information of column Description suspected of representing anomaly data.
- This data is stored in a CSV file as a reference for removing this anomaly data.
- Remove anomaly data.

```
'''
  Buang data yang bukan transaksi yang sebenarnya.
'''
totrecb4 = df.shape[0]
tot_anomaly = df[df['Description'].isin(fltr['Description'])].shape[0]

df = df[~df['Description'].isin(fltr['Description'])]

print(f'Total record before cleansing: {totrecb4}')
print(f'Total record anomaly removed : {tot_anomaly}')
print(f'Total record after cleansing : {df.shape[0]}')
```

```
Total record before cleansing: 438570
Total record anomaly removed : 318
Total record after cleansing : 438252
```

Missing Values : “Description”

```
desc_null, custid_null = df_ori['Description'].isnull().sum(), df_ori['CustomerID'].isnull().sum()
print(f'Total null data at column Description: {desc_null}')
# print(f'Jumlah data null pada kolom CustomerID: {custid_null}')
```

```
Total null data at column Description: 1454
```

```
df[df['Description'].isnull()].tail().sort_values(by='StockCode')
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
535332	581209	21620	NaN	6	12/7/2011 18:35	0.00	NaN	United Kingdom
533711	581102	21803	NaN	20	12/7/2011 11:57	0.00	NaN	United Kingdom
533712	581103	22689	NaN	4	12/7/2011 11:58	0.00	NaN	United Kingdom
535326	581203	23406	NaN	15	12/7/2011 18:31	0.00	NaN	United Kingdom
538554	581408	85175	NaN	20	12/8/2011 14:06	0.00	NaN	United Kingdom

StockCode	Description
16156L	WRAP, CAROUSEL
16156L	WRAP CAROUSEL
17107D	FLOWER FAIRY,5 SUMMER B'DRAW LINERS
17107D	FLOWER FAIRY 5 SUMMER DRAW LINERS
20681	MIA
20681	PINK POLKADOT CHILDRENS UMBRELLA
20725	LUNCH BAG RED RETROSPOT
20725	LUNCH BAG RED SPOTTY
20914	SET/5 RED SPOTTY LID GLASS BOWLS
20914	SET/5 RED RETROSPOT LID GLASS BOWLS

- There 1,454 records missing values
- Some of the same StockCode have similar, but not exactly the same Description.

Find new value of “Description”

After

```
df[df['Description'].isnull()].tail().sort_values(by='StockCode')
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	new_Description
535332	581209	21620	NaN	6	12/7/2011 18:35	0.00	null	United Kingdom	SET OF 4 ROSE BOTANICAL CANDLES
533711	581102	21803	NaN	20	12/7/2011 11:57	0.00	null	United Kingdom	CHRISTMAS TREE STAR DECORATION
533712	581103	22689	NaN	4	12/7/2011 11:58	0.00	null	United Kingdom	DOORMAT MERRY CHRISTMAS RED
535326	581203	23406	NaN	15	12/7/2011 18:31	0.00	null	United Kingdom	HOME SWEET HOME KEY HOLDER
538554	581408	85175	NaN	20	12/8/2011 14:06	0.00	null	United Kingdom	CACTI T-LIGHT CANDLES

Before

Methodology:

- Create standard reference of Description generated from data source.
- Look up the value using key: StockCode

```
# df[df['new_Description'].isnull()] #.sum()
df[df['new_Description'] == 'OTHER']
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	new_Description
1970	536545	21134	NaN	1	12/1/2010 14:32	0.00	NaN	United Kingdom	OTHER
1987	536549	85226A	NaN	1	12/1/2010 14:34	0.00	NaN	United Kingdom	OTHER
1988	536550	85044	NaN	1	12/1/2010 14:34	0.00	NaN	United Kingdom	OTHER
2024	536552	20950	NaN	1	12/1/2010 14:34	0.00	NaN	United Kingdom	OTHER
2026	536554	84670	NaN	23	12/1/2010 14:35	0.00	NaN	United Kingdom	OTHER
19625	537872	21595	NaN	1	12/8/2010 18:07	0.00	NaN	United Kingdom	OTHER
19628	537875	20849	NaN	1	12/8/2010 18:08	0.00	NaN	United Kingdom	OTHER
21782	538133	85018C	NaN	3	12/9/2010 15:56	0.00	NaN	United Kingdom	OTHER
21786	538137	62095B	NaN	2	12/9/2010 15:57	0.00	NaN	United Kingdom	OTHER
21787	538138	72814	NaN	2	12/9/2010 15:57	0.00	NaN	United Kingdom	OTHER
21791	538142	84247C	NaN	1	12/9/2010 15:58	0.00	NaN	United Kingdom	OTHER
21793	538144	90042B	NaN	1	12/9/2010 15:58	0.00	NaN	United Kingdom	OTHER
346849	567207	35592T	NaN	4	9/19/2011 11:01	0.00	NaN	United Kingdom	OTHER

Only 13 of 1,454 records failed to map, and filled by “OTHER”

Missing Values : “CustomerID”

```
df[df['CustomerID'].isnull()].head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	new_Description
1443	536544	21773	DECORATIVE ROSE BATHROOM BOTTLE	1	12/1/2010 14:32	2.51	NaN	United Kingdom	DECORATIVE ROSE BATHROOM BOTTLE
1444	536544	21774	DECORATIVE CATS BATHROOM BOTTLE	2	12/1/2010 14:32	2.51	NaN	United Kingdom	DECORATIVE CATS BATHROOM BOTTLE
1445	536544	21786	POLKADOT RAIN HAT	4	12/1/2010 14:32	0.85	NaN	United Kingdom	POLKADOT RAIN HAT
1446	536544	21787	RAIN PONCHO RETROSPOT	2	12/1/2010 14:32	1.66	NaN	United Kingdom	RAIN PONCHO RETROSPOT
1447	536544	21790	VINTAGE SNAP CARDS	9	12/1/2010 14:32	1.66	NaN	United Kingdom	VINTAGE SNAP CARDS

```
df_null = df[df['CustomerID'].isnull()]
df_null.shape
```

```
(116948, 9)
```

- Total missing value “CustomerID” = 135,080 (before data cleansing) from total data = 541,909 → 24.9%
- Because the value is needed for analysis purposes, the value of the "CustomerID" column cannot be ignored, even though the percentage is > 20%.
- Imputation using the mode approach cannot be done, because the number of missing values is quite large so it can cause bias.
- For this reason, another approach is needed to obtain the missing value of "CustomerID" by using the KNN (K-Nearest Neighbors) method.

Imputation “CustomerID”

```
Total record before cleansing       : 438252
Total null data at column CustomerID: 116948
Total null data after Imputation KNN: 0
Percentage missing value CustomerID : 26.69%
```

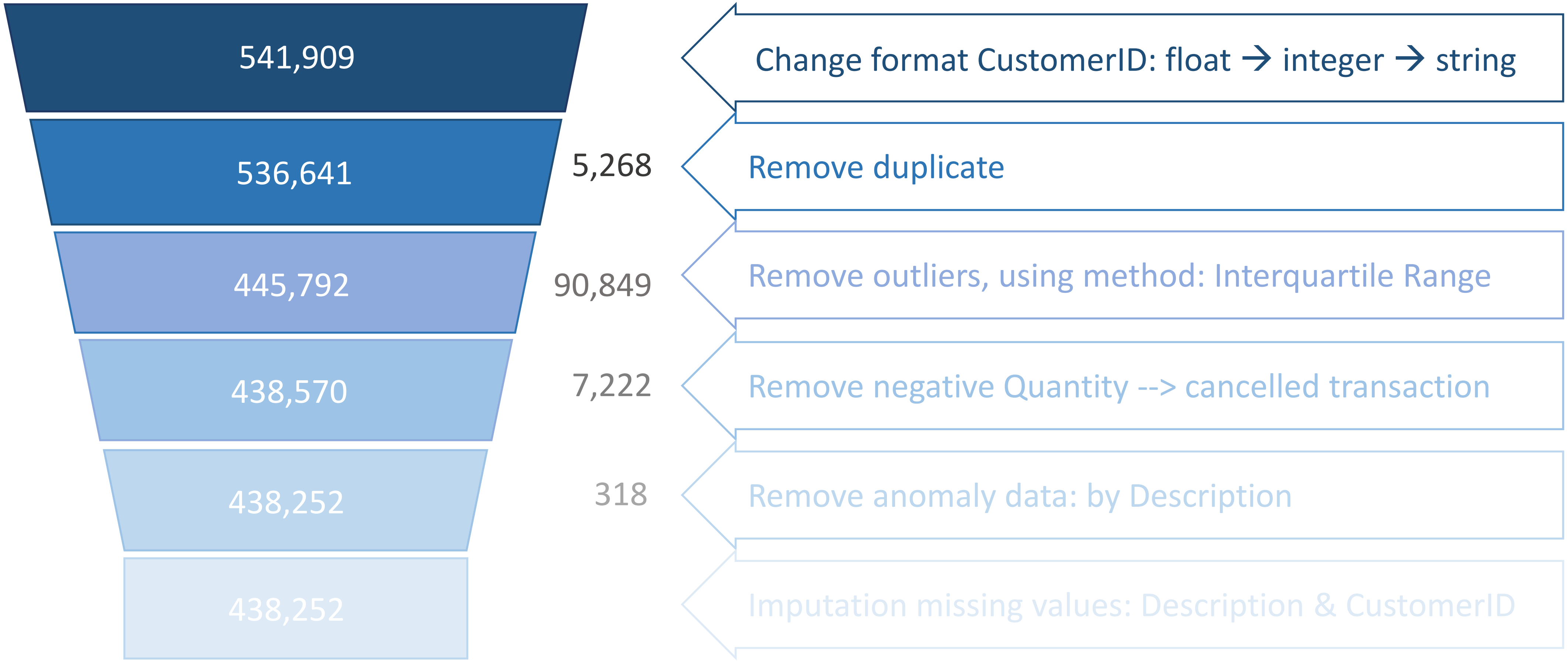
```
# Copy-kan hasil Imputasi ke df
df = df_imputed

# final check to make sure that imputation worked
# Check data yang semula CustomerID-nya null
df[df.index.isin(df_null.index)]
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	new_Description
1443	536544	21773	DECORATIVE ROSE BATHROOM BOTTLE	1	12/1/2010 14:32	2.51	14096	United Kingdom	DECORATIVE ROSE BATHROOM BOTTLE
1444	536544	21774	DECORATIVE CATS BATHROOM BOTTLE	2	12/1/2010 14:32	2.51	14096	United Kingdom	DECORATIVE CATS BATHROOM BOTTLE
1445	536544	21786	POLKADOT RAIN HAT	4	12/1/2010 14:32	0.85	12838	United Kingdom	POLKADOT RAIN HAT
1446	536544	21787	RAIN PONCHO RETROSPOT	2	12/1/2010 14:32	1.66	15311	United Kingdom	RAIN PONCHO RETROSPOT
1447	536544	21790	VINTAGE SNAP CARDS	9	12/1/2010 14:32	1.66	18109	United Kingdom	VINTAGE SNAP CARDS
...	...	...	...	...	...	...	...	...	...
541534	581498	85049A	TRADITIONAL CHRISTMAS RIBBONS	5	12/9/2011 10:26	3.29	14096	United Kingdom	TRADITIONAL CHRISTMAS RIBBONS
541535	581498	85049E	SCANDINAVIAN REDS RIBBONS	4	12/9/2011 10:26	3.29	17725	United Kingdom	SCANDINAVIAN REDS RIBBONS
541536	581498	85099B	JUMBO BAG RED RETROSPOT	5	12/9/2011 10:26	4.13	14096	United Kingdom	JUMBO BAG RED RETROSPOT
541537	581498	85099C	JUMBO BAG BAROQUE BLACK WHITE	4	12/9/2011 10:26	4.13	14096	United Kingdom	JUMBO BAG BAROQUE BLACK WHITE
541538	581498	85150	LADIES & GENTLEMEN METAL SIGN	1	12/9/2011 10:26	4.96	14096	United Kingdom	LADIES & GENTLEMEN METAL SIGN

116948 rows × 9 columns

Summary of Data Cleansing

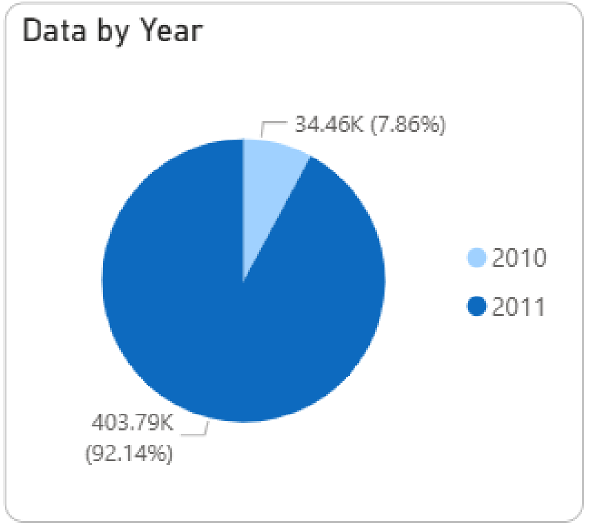
Original data set = 541,909



Data Analysis and Insights

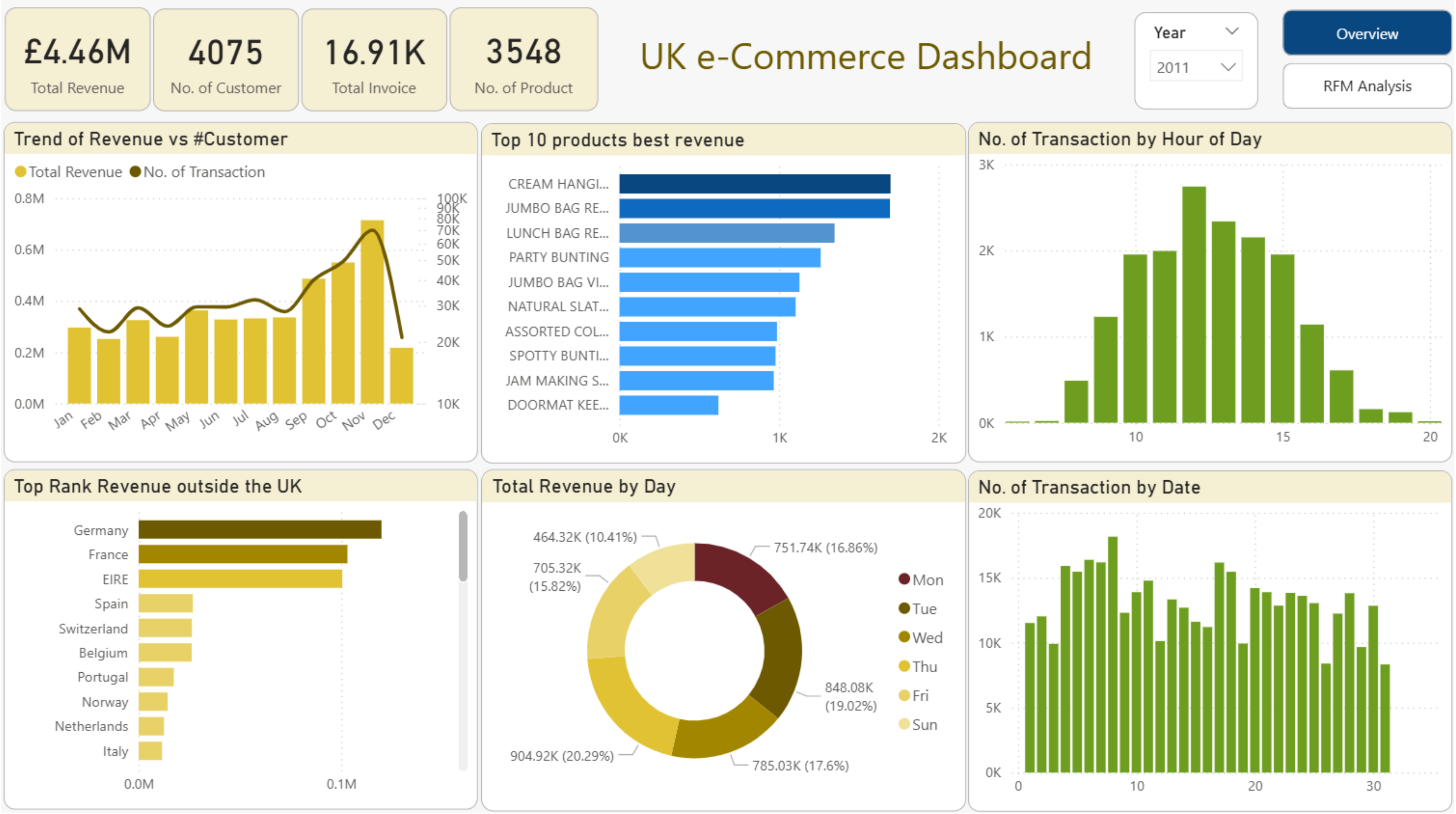
Data Completeness and Cleaness

- There are two years of data, namely: 2010 and 2011. However, the 2010 data is incomplete. So, in the next analysis only 2011 data was used.
- The data used is not clean, so it is necessary to carry out data cleansing using various methods and assumptions.
- The data cleansing carried out includes:
  1. changing data types (CustomerID) & string column case standardization (StockCode)
  2. Removing: duplicate data, outliers, canceled transactions and anomalous data
  3. data imputation by looking up to the reference table (Description) and KNN method (CustomerID).
- Of the total 541,909 rows of original data, after data cleansing, 438,252 rows of data remained, or approximately 80.87%.



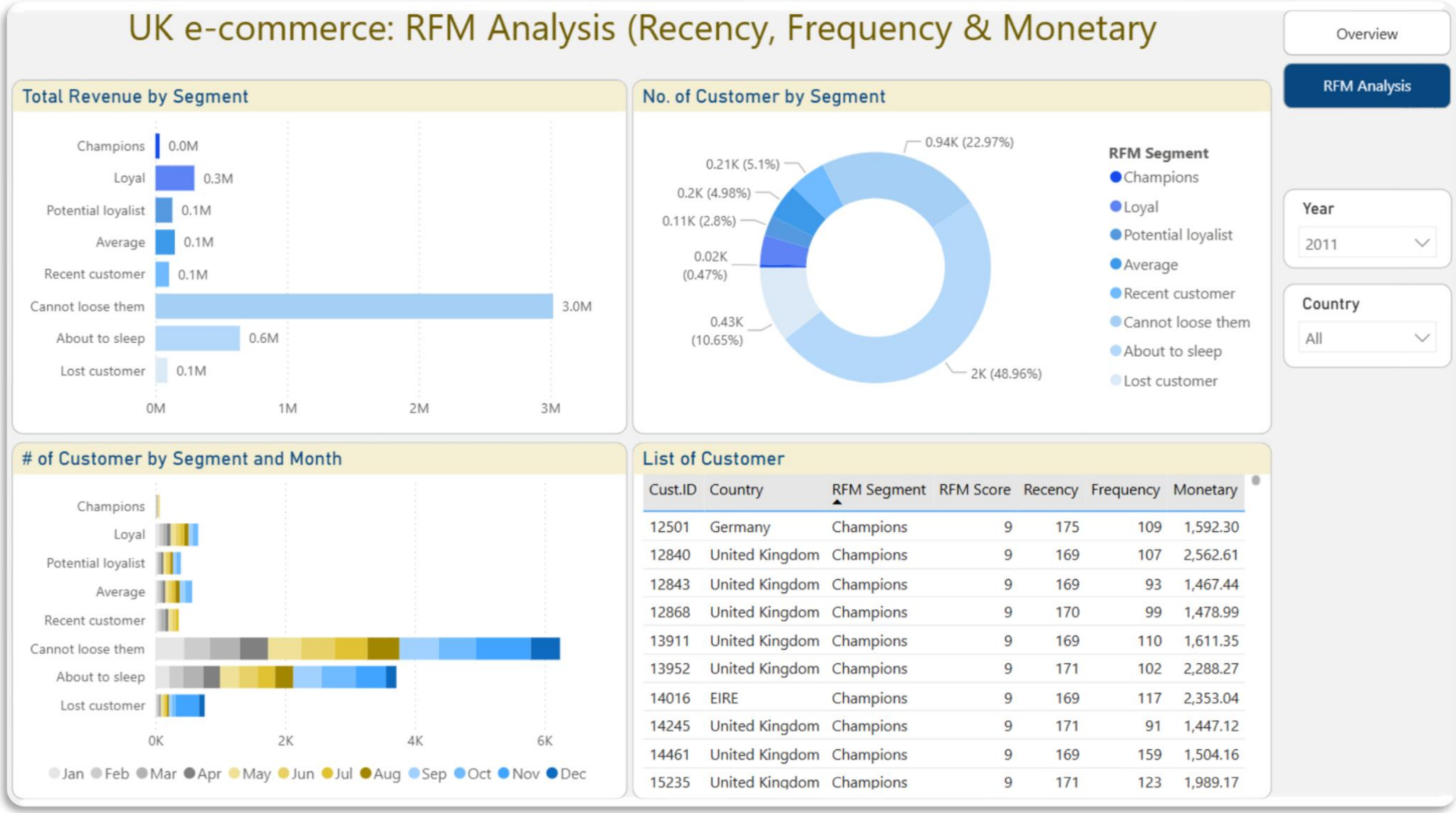
Data Visualization (Dashboard)

Overview



£4.46M
Total Revenue
4075
No. of Customer
16.91K
Total Invoice
3548
No. of Product
UK e-Commerce Dashboard
Year
2011
Overview
RFM Analysis
Trend of Revenue vs #Customer
Total Revenue
No. of Transaction
0.8M
0.6M
0.4M
0.2M
0.0M
100K
90K
80K
70K
60K
50K
40K
30K
20K
10K
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Top 10 products best revenue
CREAM HANGI...
JUMBO BAG RE...
LUNCH BAG RE...
PARTY BUNTING
JUMBO BAG VI...
NATURAL SLAT...
ASSORTED COL...
SPOTTY BUNTI...
JAM MAKING S...
DOORMAT KEE...
0K
1K
2K
No. of Transaction by Hour of Day
3K
2K
1K
0K
10
15
20
Top Rank Revenue outside the UK
Germany
France
EIRE
Spain
Switzerland
Belgium
Portugal
Norway
Netherlands
Italy
0.0M
0.1M
Total Revenue by Day
464.32K (10.41%)
751.74K (16.86%)
705.32K (15.82%)
848.08K (19.02%)
904.92K (20.29%)
785.03K (17.6%)
Mon
Tue
Wed
Thu
Fri
Sun
No. of Transaction by Date
20K
15K
10K
5K
0K
0
10
20
30

RFM Analysis



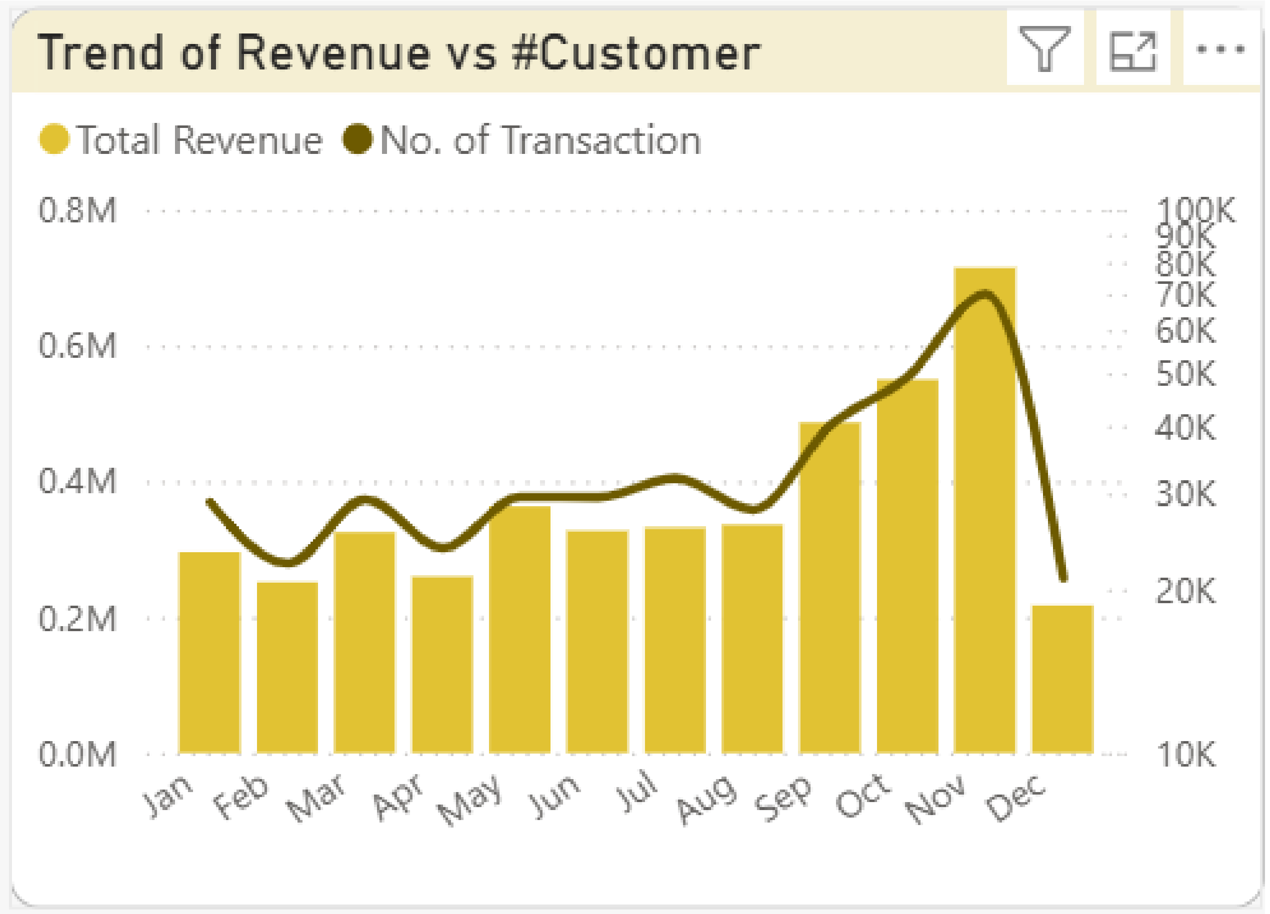
Cust.ID	Country	RFM Segment	RFM Score	Recency	Frequency	Monetary
12501	Germany	Champions	9	175	109	1,592.30
12840	United Kingdom	Champions	9	169	107	2,562.61
12843	United Kingdom	Champions	9	169	93	1,467.44
12868	United Kingdom	Champions	9	170	99	1,478.99
13911	United Kingdom	Champions	9	169	110	1,611.35
13952	United Kingdom	Champions	9	171	102	2,288.27
14016	EIRE	Champions	9	169	117	2,353.04
14245	United Kingdom	Champions	9	171	91	1,447.12
14461	United Kingdom	Champions	9	169	159	1,504.16
15235	United Kingdom	Champions	9	171	123	1,989.17

Global values

£4.46M	4075	16.91K	3548
Total Revenue	No. of Customer	Total Invoice	No. of Product

- The total revenue of e-commerce businesses is quite large and is increasing day by day, due to technological developments and changes in consumer behavior.
- The level of public trust in online shopping is increasing due to improved service, security and convenience when making transactions.
- Because of this, currently many companies are doing various ways to attract customer interest, such as: providing discounts, cashback, easy payment methods, free shipping, and even personalized product offers.
- By getting a good shopping experience, it is hoped that customers will stay and become a positive story to attract other new customers.

Trend of revenue compared to # transaction

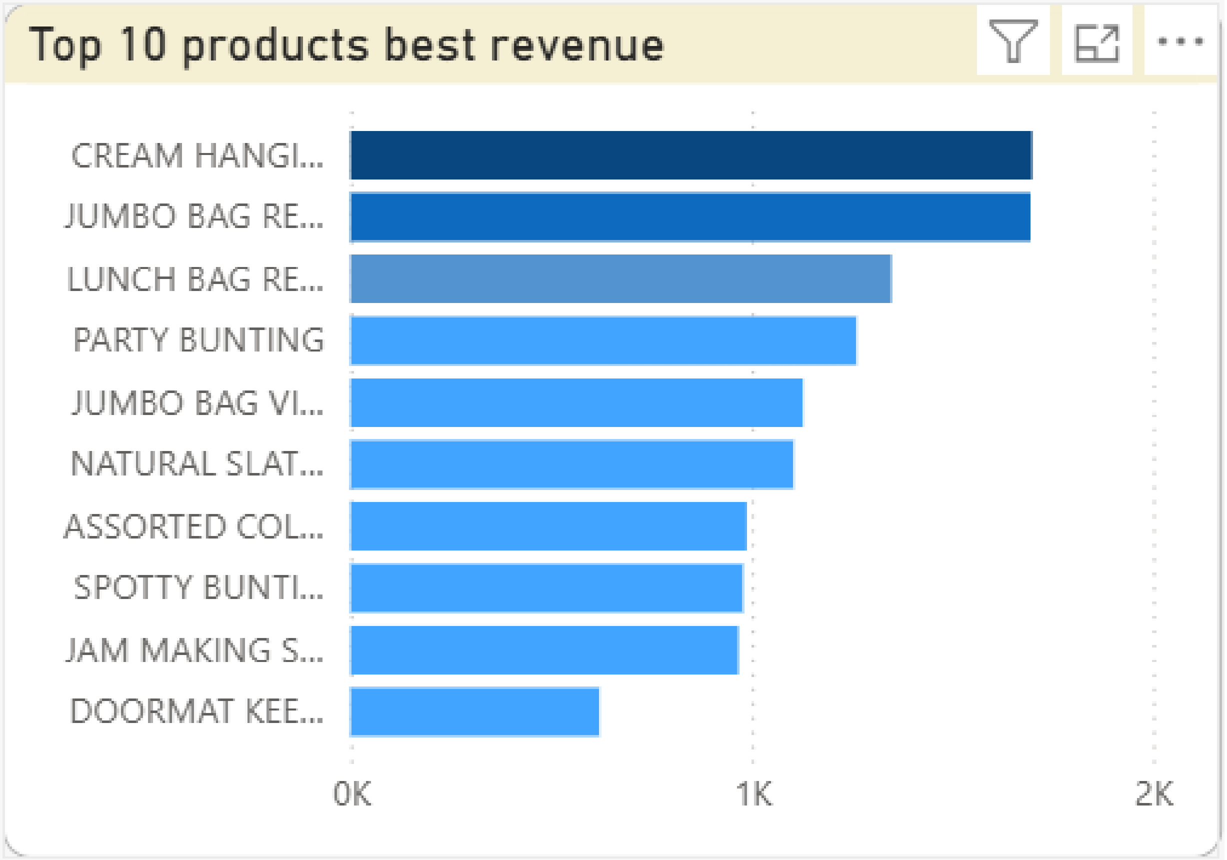


- Product sales increased significantly at the end of the year, especially in the period from September to November.
- This happened, perhaps because September started the winter season and Christmas and New Year celebrations, so that people's spending during that month increased.
- It's the right time for sellers on e-commerce platforms to provide sales incentives to customers, such as providing special discounts, bonuses or appropriate product offers.

Top rank revenue gained outside the UK

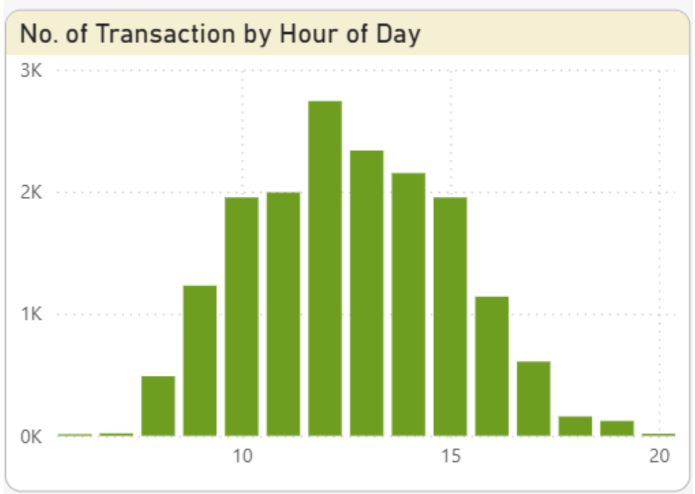
- The 10 countries outside the UK that make the biggest contribution to revenue are relatively close by, except perhaps Spain and Portugal.
- This fact can be a consideration when expanding the business. Local factors in each country need to be considered to ensure effective and efficient market penetration.

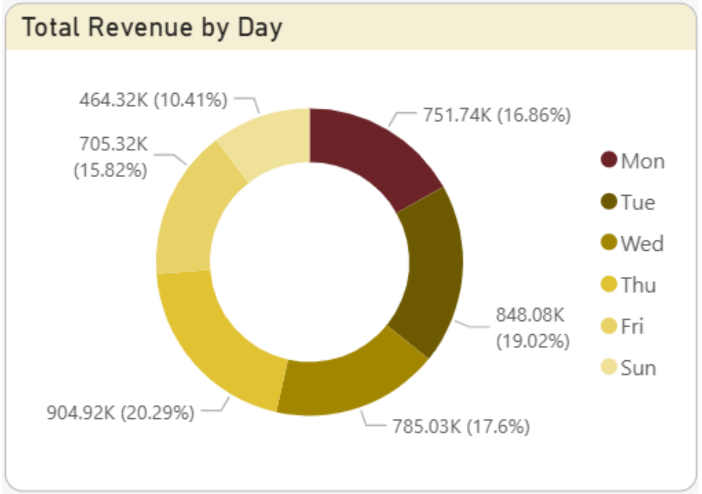
Top rank revenue gained outside the UK

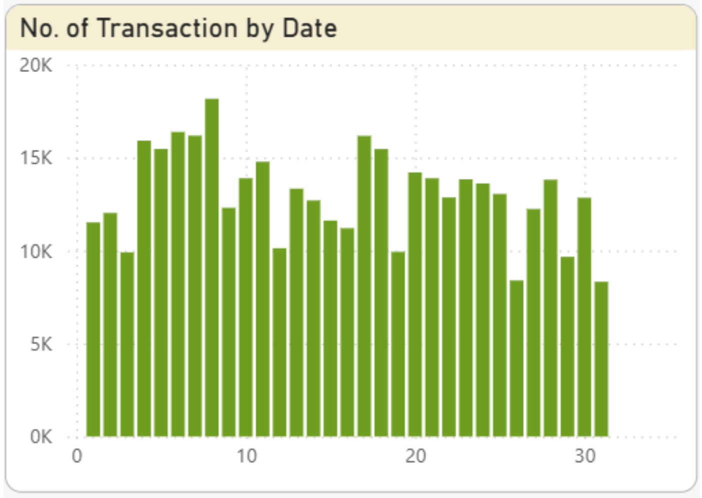


- For the most purchased top products, it's necessary to further study why these types of products sell best. It's also important to examine whether the public's demand for these products has been met.
- If possible, it's possible to import these products from countries with lower production costs, such as China, Bangladesh, or Indonesia, to meet market demand.
- Most importantly, quality control can be maintained, given that Europeans are quite concerned about quality, including copyright.

Transaction by Day/ Date & Revenue by Day





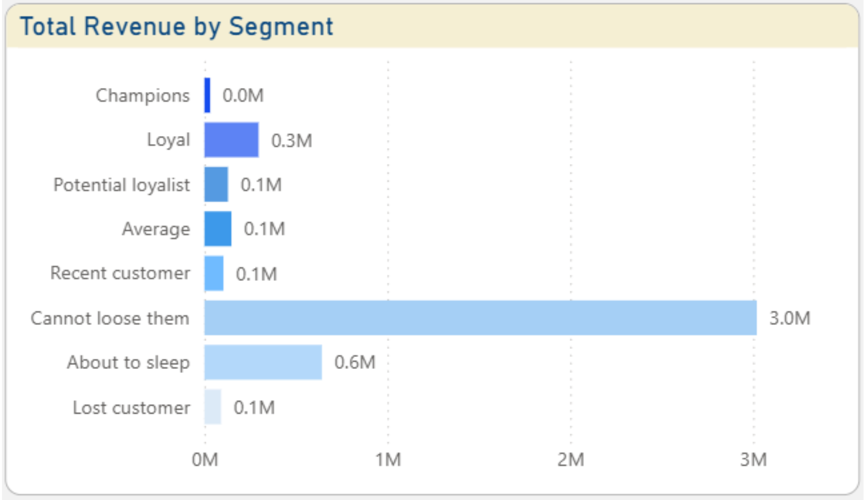


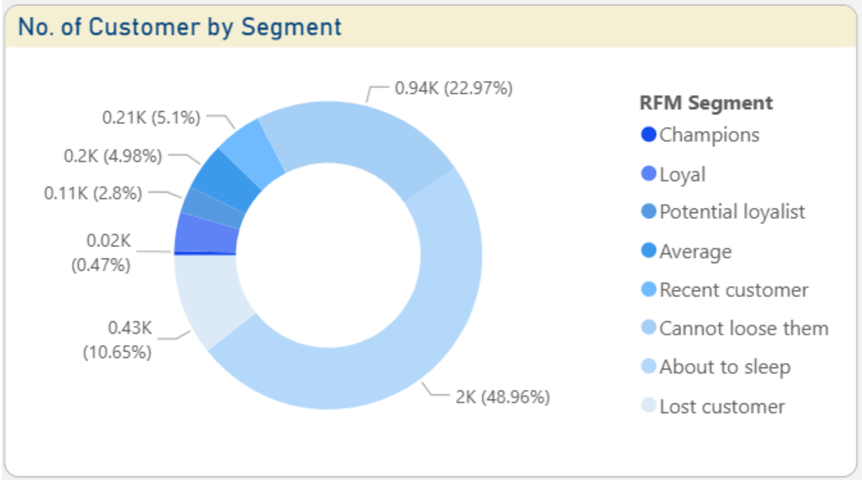
- Transactions are carried out during normal hours of the day, especially in the middle of the day when all employees are taking a break from work.
- There are no significant differences between consumer shopping days, except for Saturday, where, according to available data, there are no transactions on that day.
- People tend to shop at the beginning of the month, especially from the 1st to the 8th, perhaps because most people receive their monthly salary at the end of the month.

RFM Analysis Customer Segmentation

Segment Name	Total Score	Formula Description
Champion	9	Do transaction recently, buy often and spend the most!
Loyal	8	Bring good money for us, responsive to promotion
Potential Loyalist	7	Recent customers, but spent a good amount and bought more than once
Average	6	In average recency, frequency, and monetary values
Recent Customer	6-7 (recency days score 3)	Bought most recently, but not often
Cannot Loose Them	6-7 (recency days score 1)	Made biggest purchases, and often. But haven't returned for a long time
About to Sleep	4 - 5	Below average recency, frequency, and monetary values. Will lose them if not reactivated
Lost Customer	3	Lowest recency, frequency and monetary values

Revenue and no. of customer





- The largest spending segments come from "Cannot Lose Them" and "About to Sleep," accounting for approximately 81%. The "Lost Customer" segment accounts for 2%, contributing revenue. The remaining 17% is spread across other segments.
- In terms of number of customers, 83% of the revenue was obtained from segments with around 82.6% of customers.
- However, in terms of number of subscribers, the "About to Sleep" segment covers 48.96% of the total subscribers.

Recommendations and Actionable Insights

Champions (0.47%)

Recommendation	Actionable Insights
1. **Personalization**: Provide a personalized shopping experience by offering products relevant to their interests and preferences. 2. **Loyalty Program**: Create a loyalty program exclusive to the Champion segment, with greater and faster rewards. 3. **Effective Communication**: Ensure effective and regular communication with the Champion segment, through their preferred channels (email, SMS, or social media). 4. **Exclusive Products**: Offer exclusive or limited-edition products available only to the Champion segment. 5. **Premium Customer Service**: Provide premium and responsive customer service to ensure their satisfaction.	1. **Behavioral analysis**: Analyze the purchasing behavior of the Champion segment to understand their purchasing patterns and preferences. 2. **Further segmentation**: Further segment this Champion segment based on behavior, demographics, or preferences to enhance personalization. 3. **Product development**: Develop products that align with the needs and preferences of the Champion segment. 4. **ROI measurement**: Measure the ROI of loyalty and personalization programs to ensure the effectiveness of these investments. 5. **Retention strategy development**: Develop an effective retention strategy to retain the Champion segment and prevent them from switching to competitors.

Loyal (4.07%)

Recommendation	Actionable Insights
1. **Targeted promotions**: Offer promotions that are relevant and attractive to the Loyal segment, such as special discounts, free shipping, gifts and exclusive access. 2. **Effective communication**: Ensure effective and regular communication with the Loyal segment, through their preferred channels (email, SMS, or social media). 3. **Loyalty rewards**: Reward the Loyal segment for their loyalty, such as access to exclusive products or special events. 4. **Product development**: Develop products that meet the needs and preferences of the Loyal segment. 5. **Good customer service**: Provide good and responsive customer service to ensure their satisfaction.	1. **Behavioral analysis**: Analyze the purchasing behavior of the Loyal segment to understand their purchasing patterns and preferences. 2. **Further segmentation**: Further segment this Loyal segment based on behavior, demographics, or preferences to enhance personalization. 3. **ROI measurement**: Measure the ROI of promotions and loyalty programs to ensure the effectiveness of these investments. 4. **Retention strategy development**: Develop effective retention strategies to retain the Loyal segment and prevent them from switching to competitors. 5. **Data utilization**: Use customer data to predict the needs and preferences of the Loyal segment and offer relevant products.

Potential loyalist (2.8%)

Recommendation	Actionable Insights
1. **Effective communication**: Ensure effective and regular communication with the Potential Loyalist segment, through their preferred channels (email, SMS, or social media). 2. **Purchase rewards**: Provide rewards for their purchases, such as discounts or small gifts. 3. **Product recommendations**: Provide product recommendations relevant to their interests and preferences. 4. **Relationship development**: Develop relationships with the Potential Loyalist segment through positive and responsive interactions. 5. **Satisfaction measurement**: Measure the satisfaction of the Potential Loyalist segment to ensure they are satisfied with their purchasing experience.	1. **Behavioral analysis**: Analyze the purchasing behavior of the Potential Loyalist segment to understand their purchasing patterns and preferences. 2. **Further segmentation**: Further segment this Potential Loyalist segment based on behavior, demographics, or preferences to enhance personalization. 3. **Data utilization**: Use customer data to predict the needs and preferences of the Potential Loyalist segment and offer relevant products. 4. **Retention strategy development**: Develop an effective retention strategy to retain the Potential Loyalist segment and prevent them from switching to competitors. 5. **ROI measurement**: Measure the ROI of the retention and personalization programs to ensure the investment is effective.

Retention tactics: (1) Welcome email; (2) Special discounts; (3) Educational content; (4) Satisfaction surveys; (5) Loyalty rewards

Average (4.98%)

Recommendation	Actionable Insights
1. **Relationship development**: Develop relationships with the Average segment through positive and responsive interactions. 2. **Product recommendations**: Provide product recommendations relevant to their interests and preferences. 3. **Appropriate promotions**: Offer targeted promotions to increase purchase frequency and value. 4. **Satisfaction measurement**: Measure the satisfaction of the Average segment to ensure they are satisfied with the purchasing experience. 5. **Retention strategy development**: Develop effective retention strategies to retain the Average segment and prevent them from switching to competitors.	1. **Behavioral analysis**: Analyze the purchasing behavior of the Average segment to understand their purchasing patterns and preferences. 2. **Further segmentation**: Further segment the Average segment based on behavior, demographics, or preferences to enhance personalization. 3. **Data utilization**: Use customer data to predict the needs and preferences of the Average segment and offer relevant products. 4. **ROI measurement**: Measure the ROI of retention and personalization programs to ensure the investment is effective. 5. **Content development**: Develop relevant and engaging content for the Average segment, such as blogs, videos, or social media.

Retention tactics: (1) Email newsletter; (2) Special discounts; (3) Loyalty rewards; (4) Satisfaction surveys; (5) Loyalty program development

Recent customer (5.1%)

Recommendation	Actionable Insights
1. **Effective communication**: Ensure effective and regular communication with the Recent Customer segment, through their preferred channels (email, SMS, or social media). 2. **Purchase rewards**: Provide rewards for their purchases, such as discounts or small gifts. 3. **Product recommendations**: Provide product recommendations relevant to their interests and preferences. 4. **Relationship building**: Develop relationships with your Recent Customer segment through positive and responsive interactions. 5. **Satisfaction measurement**: Measure the satisfaction of the Recent Customer segment to ensure they are satisfied with their purchasing experience.	1. **Behavioral analysis**: Analyze the purchasing behavior of the Recent Customer segment to understand their purchasing patterns and preferences. 2. **Further segmentation**: Further segment this Recent Customer segment based on behavior, demographics, or preferences to enhance personalization. 3. **Data utilization**: Use customer data to predict the needs and preferences of the Recent Customer segment and offer relevant products. 4. **Retention strategy development**: Develop an effective retention strategy to retain the Recent Customer segment and prevent them from switching to competitors. 5. **ROI measurement**: Measure the ROI of the retention and personalization programs to ensure the investment is effective.

Retention tactics: (1) Email newsletter; (2) Special discounts; (3) Loyalty rewards; (4) Satisfaction surveys; (5) Loyalty program development

Cannot loose them (22.97%)

Recommendation	Actionable Insights
1. **Inactivity Cause Analysis**: Analyze the causes of inactivity in the Cannot Lose Them segment to understand what is preventing them from returning. 2. **Effective Communication**: Ensure effective and regular communication with the Cannot Lose Them segment, through their preferred channels (email, SMS, or social media). 3. **Loyalty Rewards**: Reward the Cannot Lose Them segment for their past loyalty. 4. **Special Offers**: Offer special and exclusive offers to win back the Cannot Lose Them segment. 5. **Retention Strategy Development**: Develop an effective retention strategy to retain and win back the Cannot Lose Them segment.	1. **Further segmentation**: Further segment the Cannot Lose Them segment based on behavior, demographics, or preferences to enhance personalization. 2. **Data utilization**: Use customer data to predict the needs and preferences of the Cannot Lose Them segment and offer relevant products. 3. **ROI measurement**: Measure the ROI of retention and personalization programs to ensure the effectiveness of these investments. 4. **Content development**: Develop relevant and engaging content for the Cannot Lose Them segment, such as blogs, videos, or social media. 5. **Satisfaction surveys**: Conduct satisfaction surveys to understand the needs and preferences of the Cannot Lose Them segment.

Retention tactics: (1) Exclusive offers; (2) Loyalty rewards; (3) Personal communication; (4) Loyalty program development; (5) Success measurement

About to sleep (48.96%)

Recommendation	Actionable Insights
1. **Customer reactivation**: Create a reactivation campaign to reactivate the About to Sleep segment by offering special offers and promotions. 2. **Effective communication**: Ensure effective and regular communication with the About to Sleep segment, through their preferred channels (email, SMS, or social media). 3. **Loyalty rewards**: Reward past loyalty to increase loyalty. 4. **Relevant offers**: Offer offers relevant to their interests and preferences to increase the likelihood of purchase. 5. **Success measurement**: Measure the success of the reactivation campaign to ensure the strategy is effective.	1. **Behavioral analysis**: Analyze the purchasing behavior of the About to Sleep segment to understand their purchasing patterns and preferences. 2. **Further segmentation**: Further segment the About to Sleep segment based on behavior, demographics, or preferences to enhance personalization. 3. **Data utilization**: Use customer data to predict the needs and preferences of the About to Sleep segment and offer relevant products. 4. **Content development**: Develop relevant and engaging content for the About to Sleep segment, such as blogs, videos, or social media. 5. **ROI measurement**: Measure the ROI of the reactivation campaign to ensure the investment is effective.

Reactivation tactics: (1) Discount offers; (2) Loyalty rewards; (3) Personal communication; (4) Loyalty program development; (5) Success measurement

Lost customer (10.65%)

Recommendation	Actionable Insights
1. **Loss Cause Analysis:** Analyze the causes of customer loss to understand what causes them to not return. 2. **Effective Communication**: Ensure effective and regular communication with the Lost Customer segment, through their preferred channels (email, SMS, or social media). 3. **Relevant Offers:** Offer offers relevant to their interests and preferences to increase the likelihood of a purchase. 4. **Loyalty Rewards**: Reward past loyalty to increase loyalty. 5. **Success Measurement:** Measure the success of win-back campaigns to ensure the strategies used are effective.	1. **Further segmentation**: Further segment the Lost Customers segment based on behavior, demographics, or preferences to enhance personalization. 2. **Data utilization**: Use customer data to predict the needs and preferences of the Lost Customers segment and offer relevant products. 3. **Content development**: Develop relevant and engaging content for the Lost Customers segment, such as blogs, videos, or social media. 4. **ROI measurement**: Measure the ROI of your win-back campaign to ensure the investment is effective. 5. **Win-back strategy development**: Develop an effective win-back strategy to retain the Lost Customers segment and increase loyalty.

Win-back tactics: (1) Discount offers; (2) Loyalty rewards; (3) Personal communication; (4) Loyalty program development; (5) Satisfaction surveys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